[SYNTHETIC PIPELINE SANITY TEST — NOT REAL LITERATURE]

Canonical Wnt Ligand Binding Inhibits GSK3-Beta to Stabilize Beta-Catenin in Osteogenesis

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Provenance: Synthetic Pipeline Sanity Test (Procedural Fixture)

Page 1 — Abstract

Bone morphogenetic pathways depend critically on canonical Wnt ligand stimulation for mesenchymal stem cell lineage allocation.

Page 2 — Introduction

Binding of Wnt3a to Frizzled receptors activates Dishevelled through cytoplasmic domain recruitment.

Page 3 — Methods & Assays

Dishevelled inhibits GSK3-beta by disrupting the axin destruction complex scaffolding.

Page 4 — Results

Inhibition of GSK3-beta prevents phosphorylation and degradation of beta-catenin, leading to cytoplasmic accumulation.

Page 5 — Discussion

Stabilized beta-catenin enters the nucleus and stimulates Runx2 transcription to promote osteoblast differentiation.

Page 6 — Conclusion

Thus, Wnt signaling suppresses GSK3-beta to stabilize beta-catenin during bone formation.